

PERSONAL STATEMENT

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A learned robot policy becomes useful only when its data, software, and hardware agree about what the robot sees and what it should do. On the CR5, I found that aligned timestamps, consistent state and action fields, and inference from a meaningful robot state could change physical behavior even when the model itself was unchanged. That experience shifted my focus from isolated prediction to the computer-science problems connecting representation, learning, serving, and physical action as operating conditions and software components change.

To gain end-to-end competence across machines and computation, I built my education as a bridge rather than a replacement. In June 2025, I earned a B.Eng. in Mechanical Design, Manufacturing and Automation at Harbin Institute of Technology, Weihai. That training grounded me in mechanism behavior, kinematics, and physical constraints. I am now pursuing a second bachelor’s degree in Computer Science and Technology at Harbin Institute of Technology, Shenzhen, expected in June 2027. This expansion lets me reason both about a machine’s physical capabilities and about how algorithms, data structures, interfaces, and runtime decisions determine its behavior. Together, these disciplines help me follow a learned capability from representation to physical execution.

My research examines how action representation shapes what a vision-language-action model can learn. As a Research Assistant at iLearn, HIT Shenzhen, I am second author of Hermite Action Tokenizer for VLA, an unsubmitted working paper, and I am responsible for its discrete autoregressive tokenization branch. I worked on splitting seven-dimensional action chunks into six-dimensional end-effector trajectories and gripper states, using shared breakpoints to remove redundant endpoints and quantization seams while coding gripper changes separately. This preserved trajectory continuity without forcing discrete gripper states into the same curve. I also compared adaptive MLP-based and fixed Hermite segmentation. This design raised a central evaluation question: how can I isolate the quality of an encoding from the behavior of a complete policy? In an offline codec benchmark averaging four LIBERO subsets with 8-by-7 action chunks, Hermite’s Endpoint MSE was approximately 99.7% lower than FAST and 82.6% lower than BEAST. The comparison demonstrates codec fidelity under that protocol; it does not establish online rollout success or physical-robot reliability.

Separate systems work on a Dobot CR5 showed how representation choices travel through an implemented learning loop. I anchored HDF5 records to RGB timestamps and aligned robot states, target actions, and gripper feedback through binary search and linear interpolation. Before downstream use, I checked schemas, timing, motion, and workspace constraints, then converted the episodes into LeRobot v2.0 with consistent seven-dimensional robot-plus-gripper states and actions. I adapted OpenPI through a training configuration, WebSocket policy serving, and a CR5 client that assembled observations, received action chunks through the server, and translated each seven-dimensional action into a six-component ServoP arm target and one Modbus gripper command. In my tests, non-blocking execution increased apparent command frequency, but a new inference could use an intermediate in-motion state before the preceding action chunk had finished, producing back-and-forth corrections. I therefore used blocking, short-horizon ServoP chunks, accepting lower apparent frequency to preserve state-action consistency. This was an execution decision rather than a change in model score. Although Hermite and CR5 were separate efforts, together they taught me to connect an algorithmic question about representation with engineering questions about clocks, software interfaces,

observations, and action execution.

Supporting projects extended this perspective. For a weld-seam HMI, I connected C++/Qt controls to Python/PyTorch segmentation inference through QProcess and visualized robot kinematics in OpenGL, keeping the desktop interface, inference process, and robot model as intelligible components. During the Xbotics Embodied AI Community Internship, I migrated an ANYmal-C navigation task from Isaac Lab into a MotrixLab NumPy environment. I refactored reset and step flows, corrected coordinate handling, and used rollout and reward/termination regression checks to examine whether the migrated environment retained its intended behavior. Both projects made cross-component change a concrete software-design problem.

My immediate goal is AI/ML or learning-systems R&D for embodied agents, building methods that remain dependable when connected to real data and hardware. I want graduate training to deepen my foundations in machine learning, data systems, and scalable software while testing them against sensing and control constraints. After gaining stronger real-system experience, I may consider doctoral study as an option rather than a predetermined step. I seek curricular depth and applied opportunities to turn this perspective into rigorous, transferable engineering. Sequential robot data are signals before they become training examples. I want to understand how alignment and preprocessing alter their temporal features, how visual observations should be modeled, and which learning methods retain useful trajectory structure. My RGB-anchored CR5 pipeline and separate action-tokenization work gave me concrete data on which to ask these questions, but not yet the signal-processing depth to answer them. NTU's MSc in Signal Processing and Machine Learning would provide that depth.

For the August 2027 intake, if the current course structure remains in place, EE6401 Advanced Digital Signal Processing would deepen how I model and transform sequential data; EE6222 Machine Vision would strengthen the visual-observation layer; and EE6406 Analytic & Ensemble Machine Learning would broaden the methods I can compare. If approved, the optional three-AU, coursework-only EE6008 Collaborative Research & Development Project could let a team vary alignment and preprocessing choices, examine their effects on temporal features, and compare how several learning methods respond. I would contribute experience with synchronized robot data and structured action representations, then carry the resulting signal-processing perspective into learning pipelines for embodied-agent R&D.